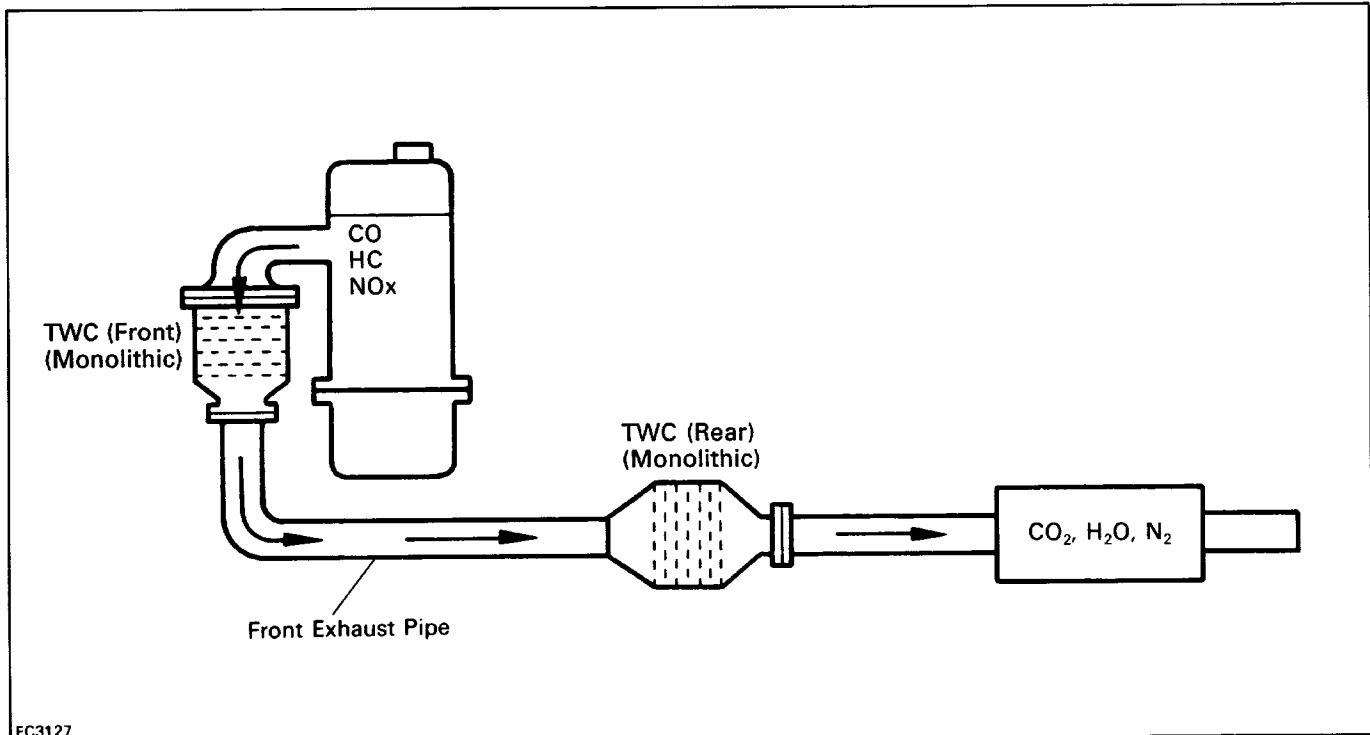


THREE-WAY CATALYTIC CONVERTER (TWC) SYSTEM

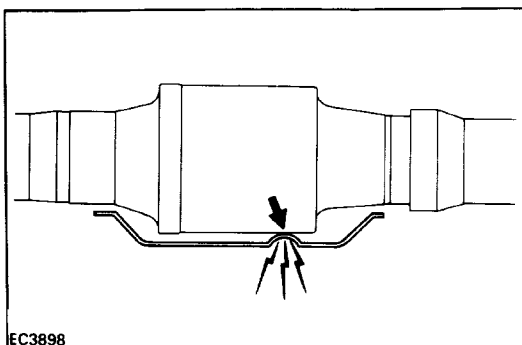


To reduce CO, HC and NO_x emissions, they are oxidized, reduced and converted to carbon dioxide (CO₂), water (H₂O) and nitrogen (N₂) by the three-way catalytic converter.

Exhaust Port		TWC (Front)		TWC (Rear)		Exhaust Gas
CO HC NO _x	→	OXIDATION AND REDUCTION	→	OXIDATION AND REDUCTION	→	CO ₂ H ₂ O N ₂

INSPECTION OF EXHAUST PIPE ASSEMBLY

1. INSPECT CONNECTIONS FOR LOOSENESS OR DAMAGE
2. INSPECT CLAMPS FOR WEAKNESS, CRACKS OR DAMAGE

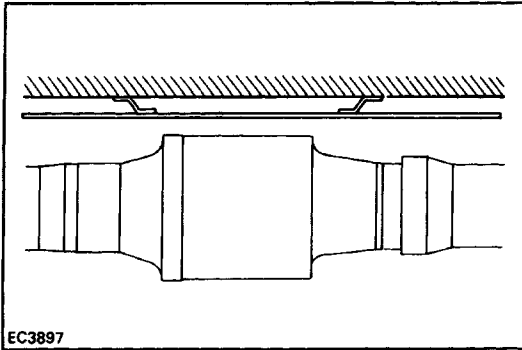


INSPECTION OF THREE-WAY CATALYTIC CONVERTER

(Three-Way Catalytic Converter-Rear)

INSPECT FOR DENTS OR DAMAGE

If any part of protector is damaged or dented to the extent that it contacts the three-way catalytic converter, repair or replace it.



INSPECTION OF HEAT INSULATOR (Three-Way Catalytic Converter–Rear)

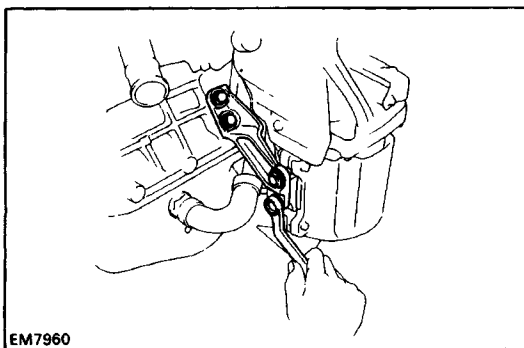
1. INSPECT HEAT INSULATOR FOR DAMAGE
2. INSPECT FOR ADEQUATE CLEARANCE BETWEEN THREE-WAY CATALYTIC CONVERTER AND HEAT INSULATOR

REPLACEMENT OF THREE-WAY CATALYTIC CONVERTER (Three-Way Catalytic Converter–Front)

1. DISCONNECT CABLE FROM NEGATIVE TERMINAL OF BATTERY

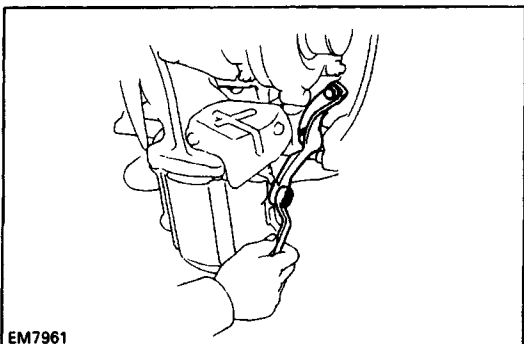
CAUTION: Work must be started after approx. 20 seconds or longer from the time the ignition switch is turned to the “LOCK” position and the negative (-) terminal cable is disconnected from the battery.

2. REMOVE ENGINE UNDER COVERS
3. REMOVE FRONT EXHAUST PIPE
(See step 2 on page [EC-30](#))
4. REMOVE GENERATOR (See page [CH-7](#))
5. REMOVE IDLER PULLEY BRACKET AND A/C COMPRESSOR WITHOUT DISCONNECTING HOSES
(See step 40 on page [EM-229](#))

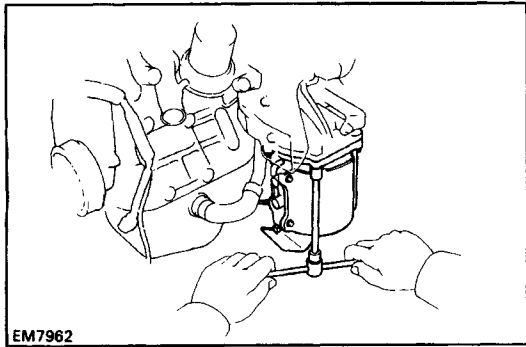


6. REMOVE THREE-WAY CATALYTIC CONVERTER (Front)

- (a) Check that the three-way catalytic converter is cool.
- (b) Remove the four bolts and RH three-way catalytic converter stay.

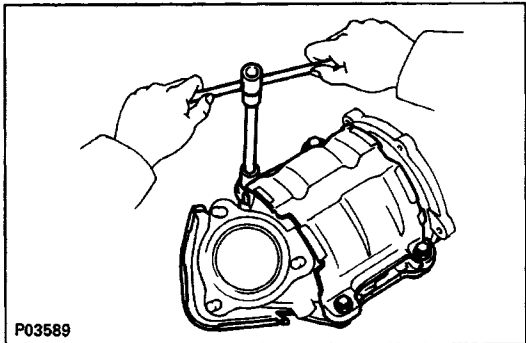


- (c) Remove the three bolts and LH three-way catalytic converter stay.



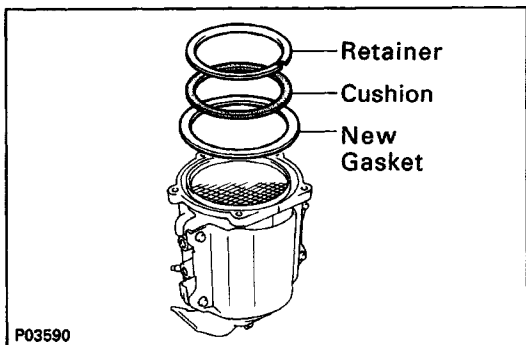
(d) Remove the three bolts, two nuts, the three-way catalytic converter, gasket, retainer and cushion.

(e) Remove the nine bolts and two heat insulators from the three-way catalytic converter.

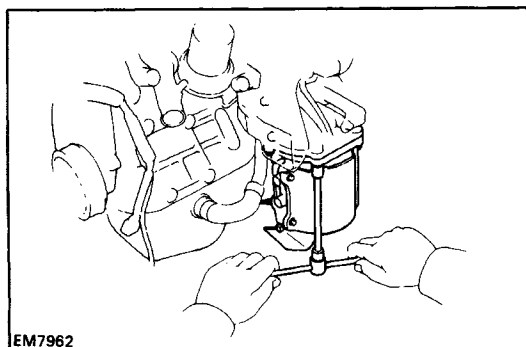


7. REINSTALL THREE-WAY CATALYTIC CONVERTER (FRONT)

(a) Install the two heat insulators to a new three-way catalytic converter with the nine bolts.

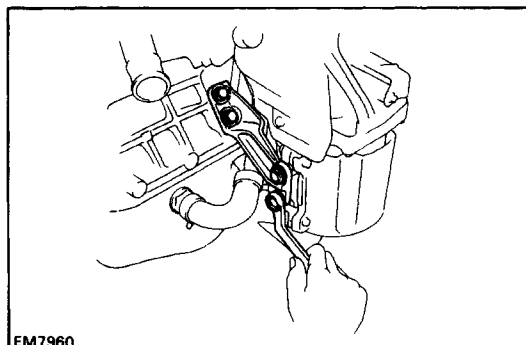


(b) Place the cushion, retainer and a new gasket on the three-way catalytic converter.



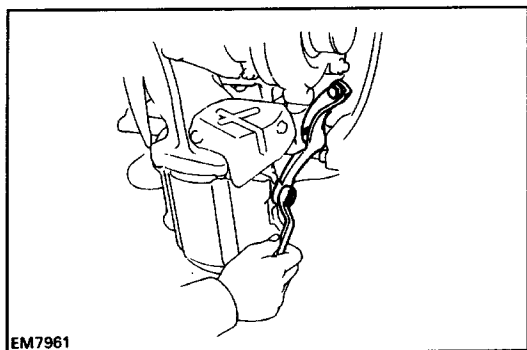
(c) Install the three-way catalytic converter with the three bolts and two new nuts.

Torque: 29 N-m (300 kgf-cm, 21 ft-lbf)



(d) Install the RH three-way catalytic converter stay with the four bolts.

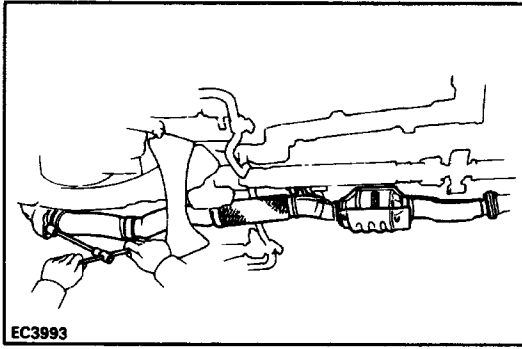
Torque: 59 N-m (600 kgf-cm, 43 ft-lbf)



(e) Install the LH three-way catalytic converter stay with the three bolts.

Torque: 59 N-m (600 kgf-cm, 43 ft-lbf)

8. REINSTALL IDLER PULLEY BRACKET AND A/C COMPRESSOR WITHOUT DISCONNECTING HOSES
(See step 12 on page [EM-261](#))
9. REINSTALL GENERATOR (See page [CH-23](#))
10. REINSTALL FRONT EXHAUST PIPE
(See step 3 on page [EC-30](#))
11. REINSTALL ENGINE UNDER COVERS
12. CONNECT CABLE TO NEGATIVE TERMINAL OF BATTERY



(Three-Way Catalytic Converter–Rear)

1. REMOVE RH ENGINE UNDER COVER

2. REMOVE FRONT EXHAUST PIPE (THREE-WAY CATALYTIC CONVERTER–REAR)

- (a) Loosen the bolt, and disconnect the clamp from the bracket.
- (b) Remove the two bolts and nuts holding the front exhaust pipe to the center exhaust pipe.
- (c) Using a 14 mm deep socket wrench, remove the three nuts holding the front exhaust pipe to the three-way catalytic converter (front).
- (d) Disconnect the support hook on the front exhaust pipe from the support bracket, and remove the front exhaust pipe and two gaskets.

3. REINSTALL FRONT EXHAUST PIPE (THREE-WAY CATALYTIC CONVERTER–REAR)

- (a) Install the support hook on the front exhaust pipe to the support bracket
- (b) Place the two new gaskets on the front and rear of the front exhaust pipe.
- (c) Temporarily install the two bolts and two new nuts holding the front exhaust pipe to the center exhaust pipe.
- (d) Using a 14 mm deep socket wrench, install the three nuts holding the front exhaust pipe to the three-way catalytic converter (front).

Torque: 62 N-m (630 kgf-cm, 46 ft-lbf)

- (e) Tighten the two bolts and nuts holding the front exhaust pipe to the center exhaust pipe.

Torque: 43 N-m (440 kgf-cm, 32 ft-lbf)

- (f) Install the clamp with the bolt.

4. REINSTALL RH ENGINE UNDER COVER